SUPPORTING ADULT GUIDANCE

Because–therefore: quadrilaterals and missing angles

Angles Tell Us Why · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, learners drag the corners of a quadrilateral (the corners snap so the four angles are whole numbers, always adding to exactly 360°) and try the special shapes: square, rectangle, parallelogram, rhombus, trapezium and kite. Drawing the diagonal shades the two triangles and shows 180° + 180° = 360°. Six reason chains follow: learners find each angle in turn and link it to its reason, building a written chain on screen. A stretch proves the 360° fact with a shaded diagram. Learners then find fourth angles, judge statements, and make quadrilaterals to order.

Driving Question: *Can a diagram prove more than it first shows?*

The mathematics, in plain terms

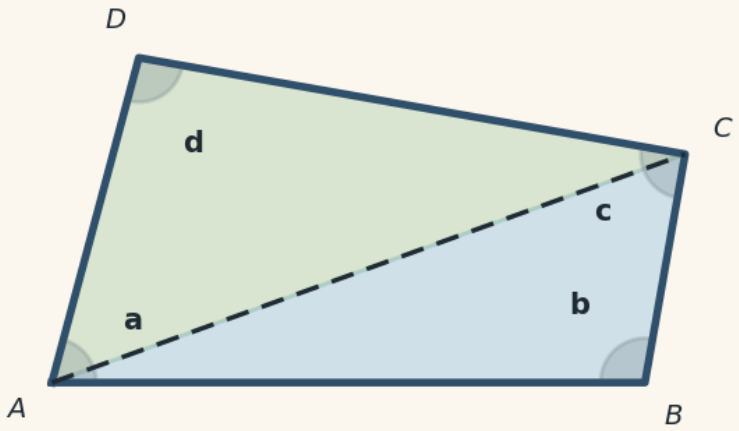
The interior angles of a (non-reflex) quadrilateral sum to 360°, since a diagonal divides it into two triangles. Parallelograms have equal opposite angles and co-interior angles summing to 180°; rectangles have four right angles; a kite has one pair of equal opposite angles (between its unequal sides). The lesson's emphasis is the chain of reasons, the form of argument used throughout secondary geometry. The general polygon formula ($(n - 2) \times 180°$) is a stretch idea only.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Why 360°?

Why do the angles a, b, c and d add up to 360°?

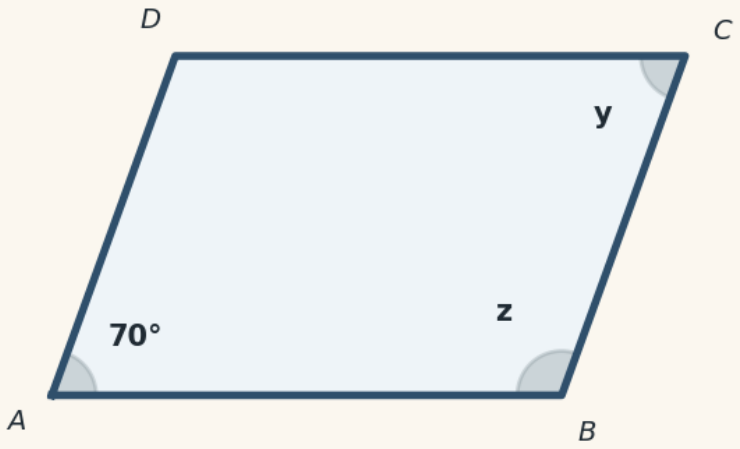


1. Draw the diagonal AC.	two triangles: ABC and ACD
2. The angles of each triangle add up to	180°
3. Together they fill the four corners exactly.	180° + 180° = 360°

a + b + c + d = 360°, for every quadrilateral.

Example 2 — A chain in a parallelogram

ABCD is a parallelogram. Find y, then z.



1. y is opposite the 70° angle.	y = 70°
2. because	opposite angles of a parallelogram are equal
3. 70 + 70 + z + z = 360, so z + z = 220.	z = 110°
4. because	angles in a quadrilateral add up to 360°

y = 70° and z = 110°: two steps, each with its reason.

See it, prove it, then chain it

Learners see the total stay at 360° and the diagonal make two triangles before any chain is built. The reason chains then require one fact per step, so learners practise writing arguments rather than just answers. The stretch states the proof in general.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The quadrilateral, the diagonal, and six reason chains. Stretch: why every quadrilateral's angles add up to 360°.	~16 min
Movement	The fourth angle: eight quick ones.	~7 min
Reflection	True or not? What 360° allows and rules out.	~7 min
Creativity	Make a quadrilateral to order.	~10 min

Rest	Big ideas built from small ones.	~2 min
Nutrition	Intellectual: argument.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using 180° for a quadrilateral.
Ask: “Draw a diagonal. How many triangles do you get?”

Giving the neighbouring angle of a parallelogram instead of the opposite one.
Ask: “Which angle is opposite? Which ones are next to each other?”

Writing answers without reasons, or with a reason that does not fit the step.
Ask: “Which fact did you use for this step?”

Mixing up which kite angles are equal.
Ask: “Which corners sit between two different-length sides?”

Trusting a measurement of a diagram as proof.
Ask: “Could the diagram be slightly wrong? What fact would make it certain?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: angles on a straight line (Lesson 2) and in a triangle (Lesson 3). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner finds z = 110° in a parallelogram but writes no reasons. The aim is a question that turns the answer into an argument.

- “How did you know y was 70°? Which fact was that?”
- “Then what did you add up, and why 360°?”

- “Can you write each step on its own line, with its reason?”
- “Would someone who only read your chain be convinced?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is the Geometry Studio used in this lesson?
Not yet: none of its current tasks fits quadrilateral angles. The lesson's own quadrilateral shows the angles exactly.

Why do the corners jump a little when dragged?
They snap so that every angle is a whole number of degrees, keeping the four labels adding up to exactly 360°.

Is the stretch required?
No. The secure core is using the 360° fact and the special quadrilaterals' angle facts in short chains of reasons.

What comes next?
This completes Angles Tell Us Why. Next: Perimeter and Area: Same Shape Story?

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.